Green-rank positivity: a master inequality and an explicit near-radial class

Working note for the "Keep the Angle" program

Purpose. Paper I reduces the rank form of Angular Preservation, for $d \le 3$, to *Green-rank positivity* (Conjecture, "Green-rank positivity"): $\rho_G(M) := \rho_S\big(d_M(X,Y), -G_M(X,Y)\big) > 0$, where G_M is the zero-mean Green kernel of the Laplace–Beltrami operator and (X,Y) are i.i.d. from the normalized volume. Two facts are known: $\rho_G = 1$ on compact two-point homogeneous spaces (Paper I, "Uniform rank preservation"), because there G is a strictly decreasing function of distance; and $\rho_G > 0$ on a C^2-open neighborhood of any metric with $\rho_G > 0$ (companion note, "Perturbative openness"). Both are qualitative and neither localizes the obstruction. This note gives (i) a *master inequality* bounding the rank deficit $1 - \rho_G$ by a single geometric quantity—the non-radiality of G, measured in sup norm against the best monotone radial profile—and (ii) an *explicit near-radial class* on which $\rho_G > 0$ with a quantitative bound. The two-point homogeneous result is the exact-radial ($R \equiv 0$) point of the master inequality, and the openness theorem is its small-R corollary made quantitative. Notation and imported facts follow Paper I and the companion reductions note (in particular Proposition "Distortion controls rank" and Lemma "Automatic atomlessness"); §5 states the standing assumptions and what remains open.

1 The master rank-remainder inequality

Fix M closed of dimension d, $\mathrm{vol}(M) = 1$, with zero-mean Green kernel G, and let (X,Y) be i.i.d. from the volume. Write $W = d_M(X,Y)$ for the random pairwise distance, $D = \operatorname{diam} M$, and let W' be an independent copy of W. All rank correlations are the population grade correlations. By Lemma "Automatic atomlessness" (companion note) the laws of $d_M(X,Y)$ and of $G(X,Y)$ are atomless (the pair law has bounded density w.r.t. $\mathrm{vol} \otimes \mathrm{vol}$), so no ties occur and Daniels' inequality $|3\tau - 2\rho_S| \le 1$ applies to every pair of the statistics below.

Definition 1. *A* monotone radial profile *is a strictly increasing continuous function* $\Psi : (0, D] \to \mathbb{R}$. *Its* remainder *is* $R(x,y) := -G(x,y) - \Psi(d_M(x,y))$, *and its* non-radiality *is* $\|R\|_\infty = \sup_{x \ne y} |R(x,y)|$. *Its* anti-concentration constant *is the essential sup of the density of the random variable* $\Psi(W)$, *denoted* $\phi_\Psi := \|f_{\Psi(W)}\|_\infty \in (0, \infty]$.

Theorem 2 (Master inequality)**.** *For every monotone radial profile* Ψ *with* $\|R\|_\infty < \infty$ *and* $\phi_\Psi < \infty$,

$$\rho_G(M) \;\ge\; 1 - 3\,\mu_\Psi\big(2\|R\|_\infty\big) \;\ge\; 1 - 12\,\phi_\Psi\,\|R\|_\infty, \qquad \mu_\Psi(\varepsilon) := \mathbb{P}\big(|\Psi(W) - \Psi(W')| \le \varepsilon\big).$$

In particular $\rho_G(M) > 0$ *whenever* $\|R\|_\infty < 1/(12\,\phi_\Psi)$*; and if* $R \equiv 0$ *(the kernel* G *is exactly a strictly decreasing function of distance), then* $\rho_G(M) = 1$*.*

Proof. Let $P = (X, Y)$ and $Q = (X', Y')$ be two independent draws of the pair, with distances d_P, d_Q and Green values G_P, G_Q. Kendall's population coefficient for the pair of statistics $(d_M, -G)$ is $\tau_G = \mathbb{E}\big[\operatorname{sgn}(d_P - d_Q)\operatorname{sgn}(G_Q - G_P)\big] = 1 - 2p$, where $p = \mathbb{P}(\text{discordant})$ and a pair-of-pairs is discordant when $\operatorname{sgn}(d_P - d_Q) \neq \operatorname{sgn}\big((-G_P) - (-G_Q)\big)$; atomlessness rules out ties. By Daniels' inequality $\rho_S \geq (3\tau_G - 1)/2 = 1 - 3p$, i.e. $\rho_G \geq 1 - 3p$.

Bound p. Suppose $d_P < d_Q$ (the reverse case is symmetric). Discordance means $-G_P > -G_Q$, i.e. $\Psi(d_P) + R_P > \Psi(d_Q) + R_Q$, whence $\Psi(d_Q) - \Psi(d_P) < R_P - R_Q \leq 2\|R\|_\infty$. Since Ψ is strictly increasing and $d_P < d_Q$, the left side is positive, so $|\Psi(d_P) - \Psi(d_Q)| \leq 2\|R\|_\infty$. As $\Psi(d_P) = \Psi(W)$ and $\Psi(d_Q) = \Psi(W')$ are independent copies, $p \leq \mathbb{P}\big(|\Psi(W) - \Psi(W')| \leq 2\|R\|_\infty\big) = \mu_\Psi(2\|R\|_\infty)$. This gives the first inequality.

For the second, let $A = \Psi(W)$ with density $f_A \leq \phi_\Psi$. Then for any $\varepsilon \geq 0$, $\mu_\Psi(\varepsilon) = \mathbb{E}_{A'}\,\mathbb{P}\big(A \in [A' - \varepsilon, A' + \varepsilon]\big) \leq \mathbb{E}_{A'}\big[2\varepsilon\,\phi_\Psi\big] = 2\phi_\Psi\varepsilon$, so $\mu_\Psi(2\|R\|_\infty) \leq 4\phi_\Psi\|R\|_\infty$ and $\rho_G \geq 1 - 12\phi_\Psi\|R\|_\infty$. Finally if $R \equiv 0$ then $p = 0$ (the reverse ranking is exact), so $\tau_G = 1$ and $\rho_G = 1$. □

Remark 3. The two known results are the two extremes. *Two-point homogeneous*: $G(x, y) = g(d_M(x, y))$ with g strictly decreasing (Paper I), so $\Psi = -g$ has $R \equiv 0$ and Theorem 2 gives $\rho_G = 1$—recovering the corollary with no separate argument. *Openness*: $\|R\|_\infty$ and ϕ_Ψ vary continuously under C^2 perturbation of the metric (the Green kernel and the pair-distance law do), so a metric with $\rho_G = 1$ (hence $R \equiv 0$ for its optimal Ψ) has a C^2-neighborhood on which $\|R\|_\infty$ stays below $1/(12\phi_\Psi)$; Theorem 2 then gives $\rho_G > 0$ *with an explicit floor*, quantifying the qualitative openness theorem. What is genuinely new is that the inequality holds on *all* metrics, not only near the radial ones: it converts Green-rank positivity into a single sup-norm estimate on the non-radial part of G.

2 The near-diagonal singularity is radial to leading order

To turn Theorem 2 into a checkable geometric statement one needs a monotone radial profile Ψ whose remainder R is finite despite the diagonal blow-up of G. The parametrix supplies one: near the diagonal G equals the Euclidean fundamental profile plus a *bounded* regular part, and the singular coefficient is universal (independent of the point), so the diagonal contributes nothing to $\|R\|_\infty$.

Let γ_d be the Euclidean fundamental profile,

$$\gamma_3(r) = \frac{1}{4\pi r}, \qquad \gamma_2(r) = -\frac{1}{2\pi}\log r,$$

(and $\gamma_d(r) = \big((d-2)\omega_{d-1}\big)^{-1} r^{2-d}$ for $d \geq 3$ in general).

Lemma 4 (Bounded regular part)**.** *Let M be a closed smooth Riemannian d-manifold, $d \in \{2, 3\}$, $\mathrm{vol}(M) = 1$, with injectivity radius $i_0 > 0$. Fix $r_0 < i_0$. There is a constant $H_0 = H_0(M, r_0) < \infty$ such that the zero-mean Green kernel satisfies*

$$\big|\,G(x, y) - \gamma_d\big(d_M(x, y)\big)\,\big| \;\leq\; H_0 \qquad \textit{whenever } d_M(x, y) \leq r_0.$$

Consequently, defining $\Psi(r) = -\gamma_d(r)$ for $r \leq r_0$ and extending Ψ to $[r_0, D]$ as any strictly increasing C^1 function with $\Psi' \geq c_1 > 0$ there and $\Psi(r_0^+) = \Psi(r_0^-)$, one has, with $H_{\text{far}} :=$ $\sup_{d_M(x,y) \geq r_0}|G(x, y)| + \sup_{r \in [r_0, D]}|\Psi(r)|$,

$$\|R\|_\infty \;\leq\; \max\big(H_0,\; H_{\text{far}}\big) \;<\; \infty,$$

and $\phi_\Psi < \infty$.

Proof. Regular part. In geodesic normal coordinates centered at y the metric is Euclidean to second order ($g^{ij} - \delta^{ij} = O(r^2)$, drift coefficients $O(r)$). Let χ be a cutoff, $\chi \equiv 1$ on $\{d_M(\cdot, y) \leq r_0\}$, supported in the i_0-ball. The parametrix $E_y(x) = \chi(x)\,\gamma_d(d_M(x, y))$ satisfies $\Delta_x E_y = \delta_y + b_y$, where $b_y = \big(\Delta_g - \Delta_{\mathrm{Eucl}}\big)\gamma_d \cdot \chi + [\Delta, \chi]\gamma_d$ collects the Euclidean-to-Riemannian discrepancy of the leading term (the $r^{2-d}/\log r$ singularity is annihilated by the Euclidean Laplacian) and the commutator with the cutoff (smooth, supported off the diagonal). Since $(\Delta_g - \Delta_{\mathrm{Eucl}})\gamma_d = O(r^{2-d})$, one has $b_y \in L^p(M)$ for every $p < \frac{d}{d-2}$ (all $p < \infty$ when $d = 2$), uniformly in y; this is where $d \leq 3$ enters. Then $h_y := G(\cdot, y) - E_y$ solves $\Delta h_y = -1 - b_y$ in the zero-mean class (from $\Delta_x G(\cdot, y) = \delta_y - 1$). Choose $p \in \big(\frac{d}{2}, \frac{d}{d-2}\big)$ —a nonempty range *exactly* for $d \in \{2, 3\}$. Elliptic $W^{2,p}$ regularity and the Sobolev embedding $W^{2,p} \hookrightarrow C^0$ on the closed manifold give $\|h_y - \bar{h}_y\|_{C^0} \leq C\|-1 - b_y\|_{L^p}$, where $\bar{h}_y = \int_M h_y = -\int_M \chi\,\gamma_d(d_M(\cdot, y))$ is bounded uniformly in y; hence $\|h_y\|_{C^0} \leq H_0$ uniformly. On $\{d_M(x, y) \leq r_0\}$, $\chi \equiv 1$ gives $E_y(x) = \gamma_d(d_M(x, y))$, so $|G(x, y) - \gamma_d(d_M(x, y))| = |h_y(x)| \leq H_0$.

Remainder bound. For $d_M(x, y) \leq r_0$, $R = -G + \gamma_d \circ d_M = -h_y$, bounded by H_0. For $d_M(x, y) \geq r_0$, both G (smooth off-diagonal, hence bounded on the compact set $\{d_M \geq r_0\}$) and $\Psi \circ d_M$ are bounded by H_{far}; hence $|R| \leq H_{\mathrm{far}}$ there.

Finiteness of ϕ_Ψ. Ψ is a strictly increasing C^1 diffeomorphism of $(0, D]$ onto its image, so $\Psi(W)$ has density $f_W(\Psi^{-1}(s))/\Psi'(\Psi^{-1}(s))$, where f_W is the density of $W = d_M(X, Y)$ (bounded, and $f_W(r) \asymp r^{d-1}$ as $r \to 0$). Near $s \to \Psi(0^+)$ ($r \to 0$), $\Psi' = -\gamma_d' \asymp r^{1-d}$ ($d \geq 3$) or r^{-1} ($d = 2$), so the density $\asymp r^{d-1}/r^{1-d} = r^{2(d-1)} \to 0$ ($d \geq 3$), resp. $r^{d-1}/r^{-1} = r^d \to 0$ ($d = 2$); on $[r_0, D]$, $\Psi' \geq c_1 > 0$ and f_W is bounded, so the density is bounded. Hence $\phi_\Psi < \infty$. □

Remark 5. The universal leading coefficient is what makes the diagonal harmless: two pairs at nearly equal small distances have Green values agreeing to leading order regardless of *where* they sit, so the singular part contributes concordantly. All the non-radiality that can hurt the ranking lives in the bounded regular part h_y (the Robin function off its radial mean) and in the mesoscopic/large-scale values H_{far}—exactly the scales at which low eigenfunctions oscillate.

3 The explicit near-radial theorem

Theorem 6 (Green-rank positivity on a near-radial class)**.** *Let M be a closed smooth Riemannian d-manifold, $d \in \{2, 3\}$, $\mathrm{vol}(M) = 1$, injectivity radius $\geq i_0$. Let Ψ_0 be the parametrix profile of Lemma 4, with finite non-radiality $\|R_0\|_\infty$ and finite anti-concentration constant ϕ_0. Then*

$$\rho_G(M) \;\geq\; 1 - 12\,\phi_0\,\|R_0\|_\infty.$$

In particular $\rho_G(M) > 0$—so, by the Green-kernel rank-limit theorem (Paper I), $\liminf_\Lambda \rho_S\big(d_M(X, Y), \|N_\Lambda(X) - N_\Lambda(Y)\|\big) = \rho_G(M) \geq 1 - 12\phi_0\|R_0\|_\infty$, the truncated commute angular ranking being uniform-in-mode-count rank-faithful with this floor—whenever

$$\phi_0\,\|R_0\|_\infty \;<\; \frac{1}{12}.$$

Writing $\Delta_{\mathrm{rad}}(M) := \inf_\Psi \|-G - \Psi \circ d_M\|_\infty$ for the best monotone-radial sup-approximation of $-G$ (the non-radiality of the Green kernel), one has $\|R_0\|_\infty \geq \Delta_{\mathrm{rad}}(M)$, so a small non-radiality with controlled ϕ_0 suffices.

Proof. Immediate from Theorem 2 applied to the *single* profile Ψ_0, whose non-radiality $\|R_0\|_\infty$ and anti-concentration constant ϕ_0 are both finite by Lemma 4; no optimization over profiles is performed (in particular the bound does *not* split the product-infimum $\inf_\Psi \phi_\Psi \|R_\Psi\|_\infty$ into separate infima—an isotonic optimal profile can be flat, giving an atom in $\Psi(W)$ and $\phi_\Psi = \infty$). The rank-faithful consequence is the Green-kernel rank-limit theorem, which gives $\rho_S \to \rho_G$; the floor is thus approached as a $\liminf$ up to $o(1)$, not attained at finite Λ. $\square$

Remark 7 (What the class contains). On the two-point homogeneous spaces $-G$ is exactly monotone-radial ($R_0 \equiv 0$, hence $\Delta_{\mathrm{rad}} = 0$ and $\rho_G = 1$); $\phi_0\|R_0\|_\infty$ is C^2-continuous, so the theorem covers a C^2-neighborhood of each with a numerical floor—the effective form of the openness theorem. More broadly, any manifold with $\phi_0\|R_0\|_\infty < 1/12$ is included: mild ellipsoids, low-eccentricity tori, small metric perturbations of symmetric spaces. Both $\|R_0\|_\infty$ and ϕ_0 are computable per metric (evaluate $-G - \Psi_0 \circ d_M$ and the density of $\Psi_0(W)$ over a volume sample), so the hypothesis is a certificate, not an article of faith. (The infimal non-radiality $\Delta_{\mathrm{rad}} \le \|R_0\|_\infty$ is a genuine geometric quantity, but it is *not* by itself sufficient: the sup-approximation optimal profile may be flat and carry $\phi = \infty$, so positivity is asserted through the fixed profile Ψ_0, whose ϕ_0 is finite.)

Remark 8 (Sharper constants via anti-concentration of $\Psi_0(W)$). The factor $12\phi_0$ is the crude one from bounding μ_{Ψ_0} by a uniform density. When the distance law is spread—$\Psi_0(W)$ far from concentrated—$\mu_{\Psi_0}(2\|R_0\|_\infty)$ is much smaller than $4\phi_0\|R_0\|_\infty$, and the first inequality of Theorem 2 ($\rho_G \ge 1 - 3\mu_{\Psi_0}$) is the honest bound to evaluate numerically. Manifolds with a well-spread pairwise-distance spectrum tolerate proportionally larger non-radiality.

4 A distributional refinement: the L^2 non-radiality bound

The master inequality controls ρ_G through $\|R\|_\infty$, the *sup* non-radiality. This is loose when $-G$ deviates from radial on only a small fraction of pairs: a short thin handle joining two distant regions makes $\|R\|_\infty$ large (the cross-handle pairs reverse) yet moves ρ_G negligibly, because those pairs are rare. The refinement below replaces the sup by the L^2 non-radiality, which is distribution-aware and, as it happens, scale-invariant—matching the fact that ρ_G is a rank correlation.

Definition 9. *For a monotone radial profile Ψ with remainder $R = -G - \Psi \circ d_M$, the L^2* non-radiality *is $V_\Psi := \mathbb{E}[R(X,Y)^2]$ and the* non-radiality index *is $\iota_\Psi := \phi_\Psi^2 V_\Psi$, with ϕ_Ψ the density bound of $\Psi(W)$, $W = d_M(X,Y)$. The index is dimensionless: ϕ_Ψ has units $[\Psi]^{-1}$ and V_Ψ units $[\Psi]^2$, so ι_Ψ is invariant under scaling the metric.*

Theorem 10 (Distributional master inequality)**.** *For every monotone radial profile Ψ with $\phi_\Psi < \infty$ and $V_\Psi < \infty$,*

$$\rho_G(M) \;\ge\; 1 - 18\left(\phi_\Psi^2 V_\Psi\right)^{1/3} \;=\; 1 - 18\,\iota_\Psi^{1/3}.$$

Hence $\rho_G(M) > 0$ whenever $\iota_\Psi < 18^{-3}$; and $V_\Psi = 0$ (radial $-G$) gives $\rho_G = 1$.

Proof. As in Theorem 2, $\rho_G \ge 1 - 3p$ with p the discordance probability, and a discordant pair-of-pairs with $d_P < d_Q$ forces $\Psi(d_Q) - \Psi(d_P) < R_P - R_Q \le |R_P| + |R_Q|$; by symmetry $p \le \mathbb{P}\big(|\Psi(d_P) - \Psi(d_Q)| < |R_P| + |R_Q|\big)$. Fix $\tau > 0$. Since $\{|\Psi(d_P) - \Psi(d_Q)| < |R_P| + |R_Q|\} \subseteq \{|\Psi(d_P) - \Psi(d_Q)| < \tau\} \cup \{|R_P| + |R_Q| \ge \tau\}$,

$$p \;\le\; \mu_\Psi(\tau) + \mathbb{P}\big(|R_P| + |R_Q| \ge \tau\big) \;\le\; 2\phi_\Psi\tau + 2\,\mathbb{P}\big(|R| \ge \tfrac{\tau}{2}\big),$$

using $\mu_\Psi(\tau) \le 2\phi_\Psi\tau$ (Theorem 2) and a union bound with R_P, R_Q i.i.d. Chebyshev gives $\mathbb{P}(|R| \ge \tau/2) \le 4V_\Psi/\tau^2$, so $p \le 2\phi_\Psi\tau + 8V_\Psi/\tau^2$ and $\rho_G \ge 1 - 6\phi_\Psi\tau - 24V_\Psi/\tau^2$. Minimizing over τ at $\tau_\star = 2(V_\Psi/\phi_\Psi)^{1/3}$ gives $6\phi_\Psi\tau_\star + 24V_\Psi/\tau_\star^2 = 18(\phi_\Psi^2 V_\Psi)^{1/3}$. □

Remark 11 (Why it beats the sup bound). Always $V_\Psi \le \|R\|_\infty^2$; the two bounds are complementary—take whichever is larger. Because of the cube-root exponent, the sup-norm master bound $1 - 12\phi_\Psi\|R\|_\infty$ is the better one in the *uniformly small* regime ($\phi_\Psi\|R\|_\infty \ll 1$), while Theorem 10 is *strictly* better exactly when the non-radiality is *concentrated*, $V_\Psi \ll \|R\|_\infty^2$. If $-G$ is non-radial by an amount $\asymp 1$ on a fraction ε of pairs and ≈ 0 elsewhere, then $\|R\|_\infty \asymp 1$ (the sup bound is vacuous) while $V_\Psi \asymp \varepsilon$, so $\iota_\Psi^{1/3} \asymp \varepsilon^{1/3} \to 0$: the L^2 bound stays sharp. This is exactly the thin-handle regime of the numerical hunt (`experiments/green-rank`): the handle spikes $\|R\|_\infty$ but leaves V_Ψ tiny, and ρ_G stays near 1—whereas a macroscopic bottleneck (a dumbbell neck) raises V_Ψ over an $\Omega(1)$ fraction of pairs and genuinely lowers ρ_G. The L^2 index is the quantity the experiments track.

Remark 12 (The constant, and the honest numerical bound). The factor 18 comes from Chebyshev and is not sharp; controlling $\mathbb{P}(|R| \ge t)$ by a higher moment or by the actual tail of R improves it, and the pre-optimization form $\rho_G \ge 1 - 6\phi_\Psi\tau - 24V_\Psi/\tau^2$ (or the master $\rho_G \ge 1 - 3\mu_\Psi(2\|R\|_\infty)$) is the one to evaluate on a sample. The value of Theorem 10 is the *quantity*: a scale-invariant, distribution-aware non-radiality index that a sup-norm cannot express.

Remark 13 (Toward a geometric class). Both factors are expected to admit geometric bounds: $\phi_\Psi \le \Phi(K, D, i_0)$ from the bounded density of the pairwise-distance law under $\mathrm{Ric} \ge -(d-1)K$, $\mathrm{diam} \le D$, $\mathrm{inj} \ge i_0$ (volume comparison), and $V_\Psi \le V(K, D, i_0, \mu_1)$ from the L^2 non-radiality of the Green kernel (its deviation from the best radial profile, controlled by heat-kernel anisotropy under the same bounds). Hence $\rho_G \ge c(K, D, i_0) > 0$ on the class $\{\iota_\Psi < 18^{-3}\}$—a strictly larger class than the sup-norm near-radial one of Theorem 6, and the one the plateau in the numerical hunt (no family below $\rho_G \approx 0.79$) suggests is broad. Making $V(K, D, i_0, \mu_1)$ explicit is the quantitative frontier item (a).

5 Standing assumptions and the frontier

Imported facts. (1) Green-function parametrix on a closed manifold: $G(x, y) = \gamma_d(d_M(x, y)) + h(x, y)$ with h bounded (indeed Hölder) up to the diagonal and γ_d the Euclidean fundamental profile—Aubin, *Nonlinear Analysis on Manifolds*, the Green-function expansion; the constant coefficient is universal. (2) Elliptic L^p regularity and Sobolev embedding on closed manifolds (Gilbarg–Trudinger; Aubin). (3) Daniels' inequality $|3\tau - 2\rho_S| \le 1$ (Daniels 1950), as used in Paper I's "Distortion controls rank". (4) Atomlessness of the pair-distance and pair-Green laws under a bounded pair density (companion note, "Automatic atomlessness"). Each is textbook or already in the program.

What is now reduced to what. Green-rank positivity is implied, with an explicit floor, by a single estimate at the fixed parametrix profile: $\phi_0\|R_0\|_\infty < 1/12$ (small non-radiality $\|R_0\|_\infty$ with controlled anti-concentration ϕ_0). The near-diagonal singularity is discharged (Lemma 4); the estimate is a statement about the *bounded* regular and large-scale parts of G only.

Open, in order of expected difficulty. (a) A geometric upper bound on $\Delta_{\mathrm{rad}}(M)$ in terms of $\mathrm{Ric} \ge -(d-1)K$, diam, i_0, and the spectral gap μ_1: the heat-kernel route ($G = \int_0^\infty (p_t - 1)\, dt$, Li–Yau two-sided Gaussian bounds on the near part, $|p_t - 1| \le Ce^{-\mu_1 t}$ on the far part) should yield

$\phi_0 \|R_0\|_\infty \le F(K, D, i_0, \mu_1)$, giving a curvature/diameter class free of the per-metric certificate. The obstacle is that the Li–Yau *constants* are not sharp, so the multiplicative near-diagonal envelope must be replaced by the sharp parametrix coefficient (Lemma 4) glued to a Li–Yau tail—routine but unpleasant. (b) Whether $\phi_0 \|R_0\|_\infty$ can exceed $1/12$ (so the sufficient condition lapses), and if so whether ρ_G can actually reach 0: the counterexample program on thin necks of revolution and Berger spheres, where a low mode oscillates strongly enough to make G badly non-radial. Theorem 2 says any such counterexample must have $\mu_\Psi(2\|R\|_\infty) \ge 1/3$ for *every* monotone profile, a concrete target for the search. (c) The critical dimension $d = 4$: the Green kernel ceases to be Hilbert–Schmidt and the rank-limit theorem's input fails; the fractional filter $\mu_k^{-s/2}$, $s > d/4$, restores it, and the first question is the analogue of Lemma 4 for the fractional kernel.